

Day 5: Troubleshooting Network Services

Practical

Task 1 :- Simulate a DNS resolution failure on your local machine by editing the hosts file to point `www.spotify.com` to a wrong IP address, then try to ping the domain and document the error message you receive.

ANS :-

1. Edit the hosts file :

Open Notepad or your text editor as Administrator :

- Windows: `C:\Windows\System32\drivers\etc\hosts`
- Linux/Mac: `sudo nano /etc/hosts`

Add this line at the bottom and save :

```
0.0.0.0 www.spotify.com
```

2. Ping the domain :

Open Command Prompt or Terminal and run :

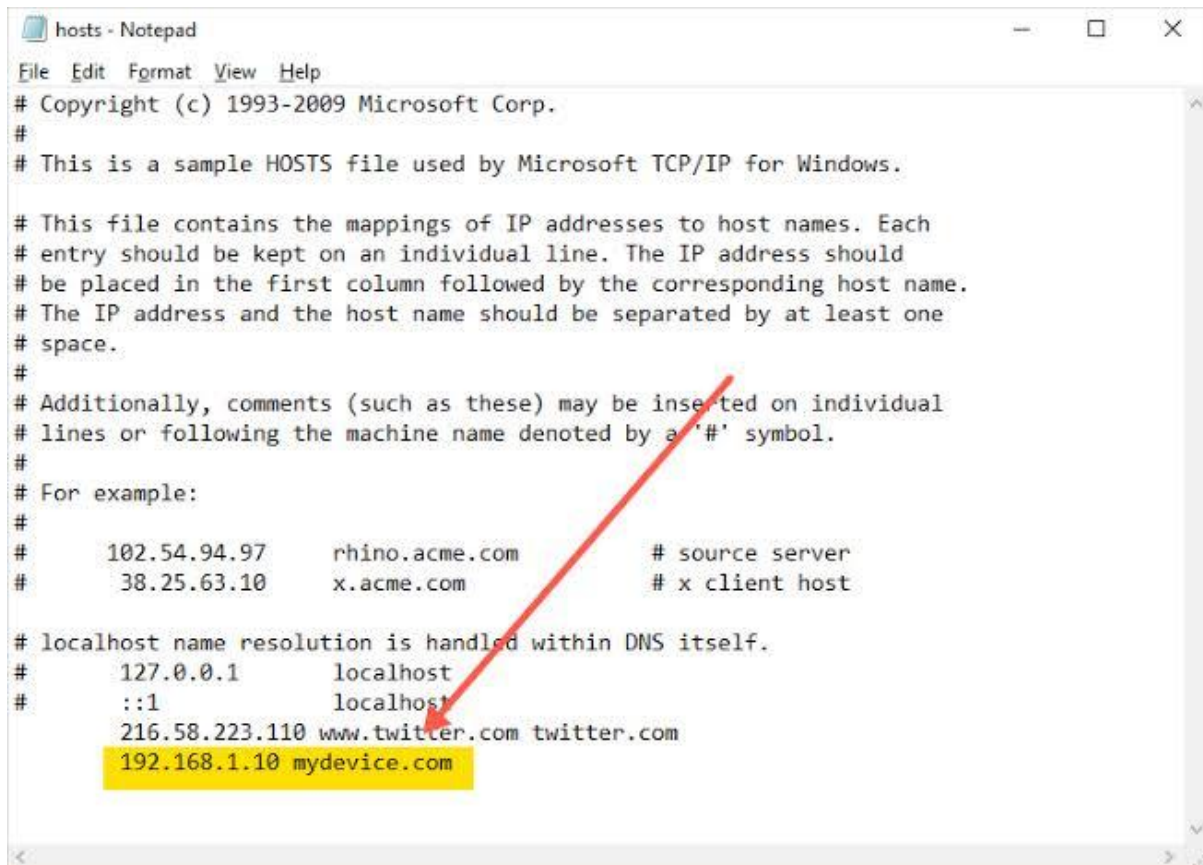
```
ping www.spotify.com
```

3.Document the error message :

You will get a connection failure because it sends traffic to the fake IP :

```
Pinging www.spotify.com [0.0.0.0] with 32 bytes of data:  
PING: transmit failed. General failure.
```

(If using an invalid local IP like 192.0.2.1, the output will be Request timed out or Destination Host Unreachable).



```
hosts - Notepad  
File Edit Format View Help  
# Copyright (c) 1993-2009 Microsoft Corp.  
#  
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.  
  
# This file contains the mappings of IP addresses to host names. Each  
# entry should be kept on an individual line. The IP address should  
# be placed in the first column followed by the corresponding host name.  
# The IP address and the host name should be separated by at least one  
# space.  
#  
# Additionally, comments (such as these) may be inserted on individual  
# lines or following the machine name denoted by a '#' symbol.  
#  
# For example:  
#  
#       102.54.94.97       rhino.acme.com       # source server  
#       38.25.63.10       x.acme.com           # x client host  
  
# localhost name resolution is handled within DNS itself.  
#       127.0.0.1         localhost  
#       ::1               localhost  
#       216.58.223.110    www.twitter.com     twitter.com  
#       192.168.1.10      mydevice.com
```

Task 2 :- Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

ANS :-

1.Configure DHCP with wrong gateway :

On your Router in Packet Tracer, configure the DHCP pool with a wrong gateway IP (192.168.1.254 instead of 192.168.1.1) :

```
ip dhcp pool MY_POOL
network 192.168.1.0 255.255.255.0
default-router 192.168.1.254
```

2.Verify on client PC :

Click the PC > IP Configuration > select DHCP.

Check the command prompt :

```
ipconfig
```

It receives the incorrect gateway: Default Gateway . . . : 192.168.1.254.

3. Identify the issue:

Try pinging an outside network :

```
ping 8.8.8.8
```

Result : Request timed out or Destination Host Unreachable because traffic is sent to an invalid gateway IP.

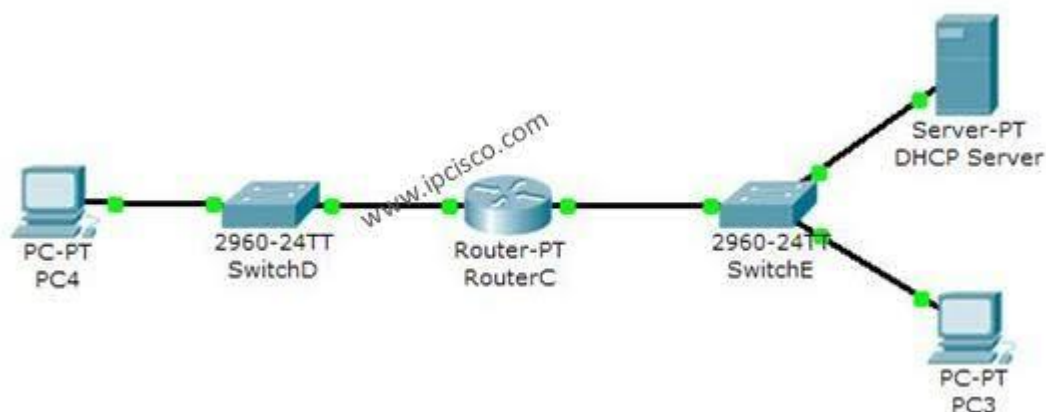
4. Fix the configuration:

Update the DHCP pool on the router with the correct router IP (192.168.1.1):

```
ip dhcp pool MY_POOL  
default-router 192.168.1.1
```

On PC command prompt, refresh IP settings :

```
ipconfig /renew
```



Task 3 :- Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

ANS :-

1.Test default DNS lookup :

Run the command :

```
nslookup google.com
```

Result : DNS request timed out or Server failed. This proves your PC's current DNS server is unreachable or misconfigured.

2.Test with a working public DNS server :

Query Google's public DNS (8.8.8.8) directly :

```
nslookup google.com 8.8.8.8
```

Result : It successfully returns the IP address (142.250.x.x). This proves internet access is working fine, but your assigned local DNS IP is bad.

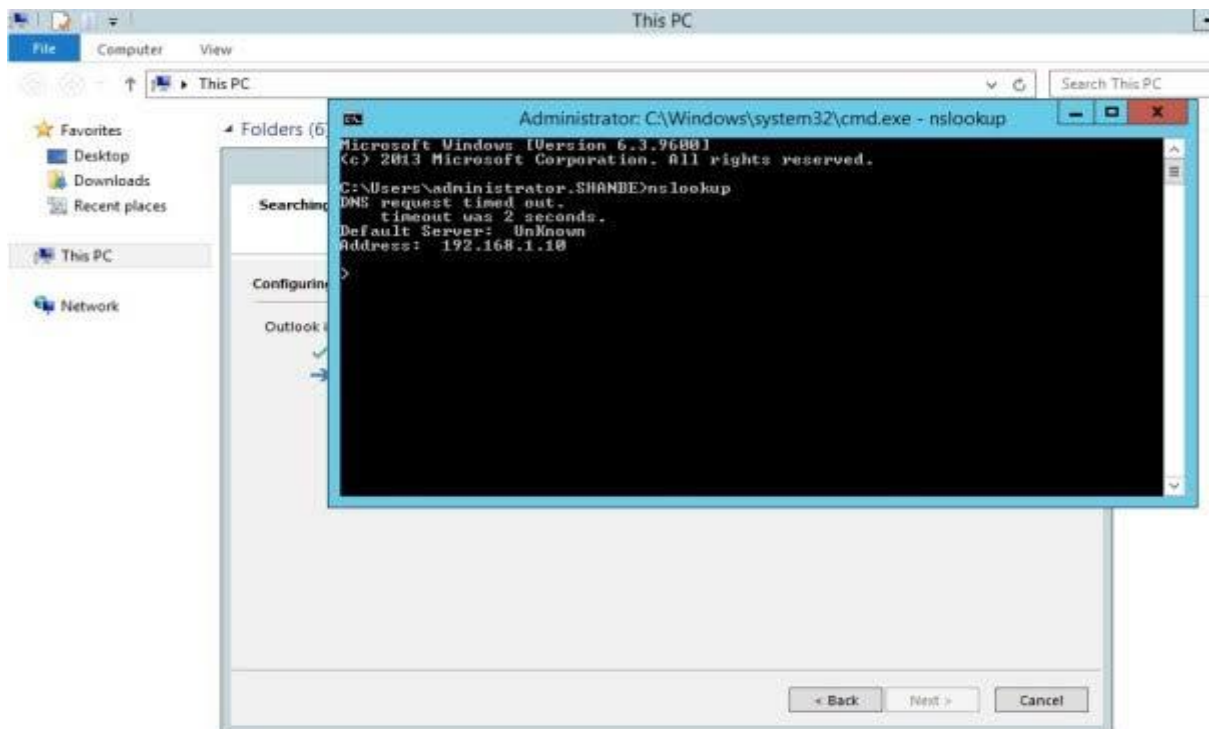
3. Apply the solution:

Change your network adapter or router's DNS settings to a reliable DNS server:

- **Primary DNS:** 8.8.8.8 (Google)
- **Secondary DNS:** 1.1.1.1 (Cloudflare)

Flush the local cache:

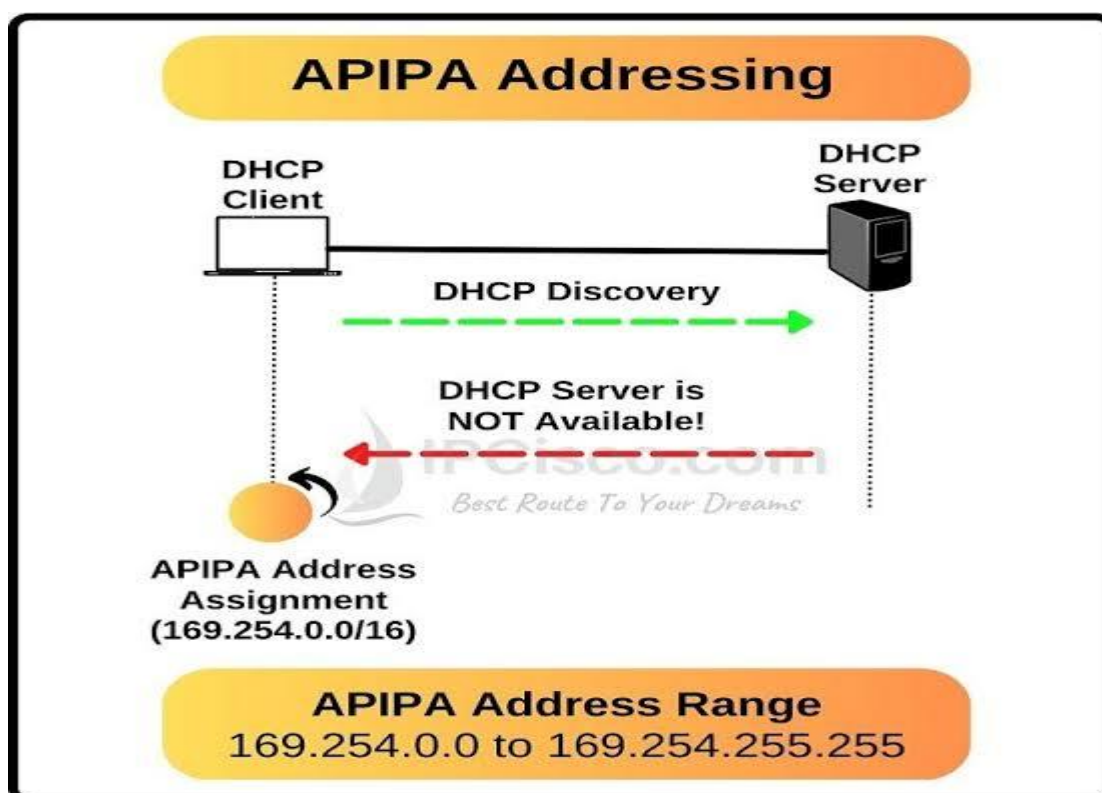
```
ipconfig /flushdns
```



Task 4 :- A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it.

ANS :-

An APIPA address (169.254.x.x) means your computer tried to ask for an IP, but the DHCP server never responded.



Possible Causes :-

- DHCP server is offline or its service stopped running.
- Network connection broken (disconnected cable, wrong VLAN, or switch port disabled).
- DHCP pool exhausted (no free IP addresses left to assign).
- Firewall blocking DHCP traffic (UDP ports 67/68).

1.Check physical connection:

Verify the cable is connected or the switch port is active and assigned to the correct VLAN.

2.Check DHCP server status:

Log into the DHCP server/router and verify the service is running and has available IPs in its pool.

3.Release and renew IP on client:

Force the device to request a new IP address:

```
ipconfig /release  
ipconfig /renew
```

4.Verify assigned IP:

Check if a normal IP (like 192.168.1.x) was received:

```
ipconfig
```

Task 5 :- Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

ANS :-

Issue	Symptoms	Possible Causes	Recommended Solution
1. APIPA Assigned	PC gets 169.254.x.x IP and has no internet access.	DHCP server service is down or network cable disconnected.	Restart DHCP server service (systemctl restart isc-dhcp-server) and check cable/VLAN.
2. IP Address Conflict	Error pop-up: <i>"An IP address conflict was detected."</i>	Static IP assigned to a device overlaps with DHCP pool.	Exclude static IPs from the DHCP pool or run ipconfig /renew.
3. No Gateway Assigned	Devices get an IP address but cannot reach outside networks.	default-router missing or mistyped in DHCP pool settings.	Edit DHCP configuration pool and set the correct gateway IP address.